Problem H2: Electroweak Mixing of $W^{(1,2,3)}$ and B

Glashow–Salam–Weinberg Unification

Parker — PHYS 7502

May 2, 2026

Problem Statement

H2. Give the relation that describes the "mixing" of the "fundamental" W_0 (this means the $\{W_1, W_2, W_3\}$ vector boson triplet) and B vector bosons of the electroweak theory, which get transformed in terms of the weak mixing angle into the usual 4 electroweak bosons we observe (W^+, W^-, Z^0, γ).
(This is related to the "Electroweak Unification" part of Griffiths.)

Deliverables

1. State the gauge group whose structure produces the four "fundamental" bosons $\{W_1, W_2, W_3, B\}$.
2. Give the relation between the charged physical bosons $W_\mu^\pm$ and the $W_\mu^{(1,2)}$ triplet members.
3. Give the rotation relation between the neutral physical bosons $\{A_\mu, Z_\mu\}$ and the neutral fundamental bosons $\{W_\mu^{(3)}, B_\mu\}$, expressed in terms of the weak mixing (Weinberg) angle θ_W.
4. Give the definition of θ_W in terms of the two underlying gauge couplings g_W and g', and the consistency relation between g_W, g', and the electromagnetic coupling g_e.
5. Briefly explain *why* the mixing happens (electroweak symmetry breaking).

Background & References

Primary references for this problem.

- **Griffiths** (2nd ed.), *Introduction to Elementary Particles*, **Ch. 9 "Weak Interactions"**, specifically:
  - §9.6 "Neutral Weak Interactions"
  - §9.7 "Electroweak Unification"
    - §9.7.1 "Chiral Fermion States"
    - §9.7.2 "Weak Isospin and Hypercharge"
    - §9.7.3 "Electroweak Mixing" ← this is the section

  Key equations: **Eq. 9.148** ($W^{\pm}$ definition), **Eq. 9.151** (the neutral mixing matrix), **Eq. 9.154** (the coupling consistency relation).
- **Thomson**, *Modern Particle Physics*, Ch. 15 "Electroweak Unification" (§15.2–15.3); Eqs. 15.12, 15.17–15.18, 15.30.
- **Halzen & Martin**, *Quarks and Leptons*, Ch. 13 (Gauge Symmetries) and Ch. 14–15 (The Weinberg–Salam Model).
- **Mann**, *An Introduction to Particle Physics and the Standard Model*, Ch. 22 (Electroweak Unification), Ch. 23 (Electroweak Symmetry Breaking).
- **Lancaster & Blundell**, *QFT for the Gifted Amateur*, Ch. 47 (The Weinberg–Salam Model).

Convention note. Griffiths conventions take priority. Mann uses opposite signs in two places (the $W^{\pm}$ definition and the inverse rotation direction) — these are flagged when they appear.

Intuition

Roadmap for this problem. The question asks for a single relation but it has more moving parts than it looks. Here's the path we'll take in five parts:

- **Part 1.** Where do the four "fundamental" bosons $\{W^1, W^2, W^3, B\}$ even come from? (Answer: they are the gauge fields of $SU(2)_L \otimes U(1)_Y$.)
- **Part 2.** How do W^1 and W^2 combine into the charged $W^{\pm}$? (Pure $SU(2)$ structure; *no* mixing angle yet.)
- **Part 3.** How do W^3 and B rotate into γ and Z? (*This* is where θ_W enters — it's the heart of the question.)
- **Part 4.** What fixes the value of θ_W? (Demanding that one of the rotated states is the QED photon.)
- **Part 5.** *Why* does the rotation happen at all? (Higgs mechanism breaks $SU(2)_L \otimes U(1)_Y \to U(1)_{\text{em}}$.)

Each part has a strategy section, slow step-by-step derivation, and explicit physical intuition.

Part 1 — Where do $W_\mu^{1,2,3}$ and B_μ come from?

Problem Statement

Restatement. Before we can "mix" anything, we have to know where the four fundamental bosons live. This part identifies the gauge group whose generators *produce* these four gauge fields, and specifies what charge each field couples to.

Deliverables

1. Name the gauge group of the electroweak interaction and explain the meaning of each factor.
2. Show that this group structure forces *exactly four* gauge bosons.
3. Identify which boson couples to which conserved charge.
4. State the Gell-Mann–Nishijima relation $Q = T^{(3)} + Y/2$, since it underlies *everything* in the rest of the problem.

Strategy

There are several ways to motivate the gauge group. The cleanest pedagogical path is:

- **Approach A (analogy with QED, the slow safe path):** Recall how QED is built as a $U(1)$ gauge theory — one symmetry, one gauge boson, one charge. Then *enlarge* the symmetry to capture the experimentally-observed weak interaction structure.
- **Approach B (counting generators, the quick path):** Note that $SU(N)$ has $N^2 - 1$ generators and $U(1)$ has 1. Pick the smallest non-trivial group that gives more than one boson and contains a piece that survives as $U(1)_{\text{em}}$.
- **Approach C (empirical, the physical path):** List the experimental facts the theory must explain (V-A charged currents, neutral currents, parity violation, massless photon, two heavy weak bosons). Show that $SU(2)_L \otimes U(1)_Y$ is the minimal symmetry consistent with all of them.

We'll do Approach A in detail and weave in pieces of B and C as we go.

1.1 — Refresher: how does QED arise as a gauge theory?

Term Refresher

Gauge symmetry. A symmetry where the transformation parameter is allowed to depend on spacetime: $\psi(x) \to e^{i\alpha(x)}\psi(x)$. Demanding the Lagrangian be invariant under this *forces* you to introduce a gauge field A_μ that transforms as $A_\mu \to A_\mu - \frac{1}{q}\partial_\mu\alpha(x)$, and the Lagrangian must use the covariant derivative $D_\mu = \partial_\mu + iqA_\mu$ instead of ∂_μ. The number of gauge fields equals the number of independent gauge parameters — which is the number of *generators* of the symmetry group.

Generator. The Hermitian operator T such that $U = e^{i\alpha T}$ is a group element. For a Lie group of dimension d, there are d independent generators.

Counting rule. For a compact Lie group G, the number of gauge bosons equals $\dim(G)$:

- $U(1)$: $\dim = 1$, gives 1 boson (the photon, in QED).
- $SU(N)$: $\dim = N^2 - 1$. So $SU(2)$ gives 3 bosons, $SU(3)$ gives 8, etc.
- For a product group $G_1 \otimes G_2$: number of bosons is $\dim(G_1) + \dim(G_2)$.

QED as the prototype. QED is built on the gauge group $U(1)_{\text{em}}$. The local phase rotation $\psi(x) \to e^{iQ\alpha(x)}\psi(x)$ (where Q is the electric charge of ψ) is the symmetry. There is *one* generator (the charge Q), so there is *one* gauge boson: the photon A_μ. Its coupling strength is g_e (the unit electric charge), and the QED interaction Lagrangian is

$$\mathcal{L}^{\text{int}}_{\text{QED}} \;=\; -\,g_e\, j^{\text{em}}_\mu A^\mu, \qquad j^{\text{em}}_\mu \;=\; \sum_f Q_f\, \bar{\psi}_f \gamma_\mu \psi_f. \tag{1}$$

This is the template we want to generalize.

1.2 — Empirical clues that point to $SU(2)_L \otimes U(1)_Y$

Physical Insight

What did experimentalists *know* about the weak interaction by the early 1960s, before GWS? Five facts:

1. The charged-current weak interaction has a $V - A$ (vector minus axial-vector) structure: it couples *only* to left-handed fermions and right-handed antifermions. (Wu's ^{60}Co experiment, 1957.)
2. Charged weak currents are mediated by some massive charged boson(s) (mass scale set by the Fermi constant G_F).
3. The known fundamental fermions sit naturally in pairs — the doublets $\binom{\nu_e}{e^-}_L$, $\binom{u}{d}_L$, etc.
4. The photon is massless, parity-conserving, and couples to electric charge.
5. There must (probably) also be a neutral-current weak interaction, whose existence was *predicted* by GWS and confirmed at Gargamelle in 1973.

Items (1), (3) point to a chiral $SU(2)$ that acts only on left-handed doublets — **this is $\boldsymbol{SU(2)_L}$**. Item (4) demands a separate $U(1)$ with a charge that *survives* as electric charge after symmetry breaking — this is what becomes $U(1)_{\text{em}}$, but *before* symmetry breaking it is the larger $\boldsymbol{U(1)_Y}$ of weak hypercharge. The minimal gauge group consistent with all five facts is therefore $SU(2)_L \otimes U(1)_Y$.

Concept / Definition

The electroweak gauge group.

$$\boxed{G_{\text{EW}} \;=\; SU(2)_L \otimes U(1)_Y}$$

The subscripts mean:

- "L" on $SU(2)_L$: this group acts only on *left-handed* fermion doublets. Right-handed fermions are $SU(2)_L$ singlets and don't feel it. This is what encodes V-A / parity violation right into the structure of the gauge group.
- "Y" on $U(1)_Y$: this is *not* the QED $U(1)_{\text{em}}$. It is the $U(1)$ of *weak hypercharge* Y. Both chiralities feel it (it's not chiral), but with different Y values for left- and right-handed fields.
- The $\otimes$ is a direct product: the two factors commute, are independent, and *neither* is contained in the other.

1.3 — Counting the gauge bosons

Step

Count generators of each factor:

- $SU(2)_L$: dim $= 2^2 - 1 = 3$. Three generators $T^a = \frac{1}{2}\sigma^a$ $(a = 1, 2, 3)$ where σ^a are the Pauli matrices acting on the left-handed weak-isospin doublet. $\Rightarrow$ **three gauge fields** W^1_μ, W^2_μ, W^3_μ, all with the *same* coupling g_W (universality is forced by non-Abelian gauge invariance).
- $U(1)_Y$: dim $= 1$. One generator (the hypercharge operator Y). $\Rightarrow$ **one gauge field** B_μ, with coupling g' (Griffiths notation; sometimes written g_Y).

Total: $3 + 1 =$ $\boxed{\text{4 fundamental gauge bosons}}$. This matches the four physical bosons $\{W^+, W^-, Z^0, \gamma\}$, as required.

Intuition

Why universality? In QED you can scale the photon coupling by hand: pick any g_e you like for each species of fermion (electrons, muons, quarks). In a non-Abelian gauge theory like $SU(2)_L$, that freedom is gone — the structure constants ϵ^{abc} in the commutator $[T^a, T^b] = i\epsilon^{abc}T^c$ *tie* the three couplings together. All three of W^1, W^2, W^3 couple with the same strength g_W.

The $U(1)_Y$ coupling g' is independent of g_W because the two factor groups don't talk to each other. This is exactly why we end up with two free coupling constants in the electroweak Lagrangian — and why $\theta_W \equiv \arctan(g'/g_W)$ is non-trivial.

1.4 — What does each boson couple to?

Concept / Definition

The interaction Lagrangian for the fermion sector is

$$\mathcal{L}_{\text{int}} \;=\; -\,g_W\,\vec{j}_\mu \cdot \vec{W}^\mu \;-\; \tfrac{g'}{2}\,j^Y_\mu B^\mu, \tag{2}$$

where:

- $\vec{j}_\mu = (j^1_\mu, j^2_\mu, j^3_\mu)$ is the **weak isospin current**, an $SU(2)$ vector built from $\bar{\chi}_L \gamma_\mu T^a \chi_L$, where χ_L is a left-handed doublet. This couples to the triplet $\vec{W}_\mu$.
- $j^Y_\mu = \sum_f Y_f\,\bar{\psi}_f \gamma_\mu \psi_f$ is the **weak hypercharge current**, a singlet, summed over both chiralities (with potentially different Y for left- and right-handed components). This couples to B_μ.
- The factor of $\frac{1}{2}$ in front of g' is purely a Griffiths convention — it's there so that hypercharge enters the covariant derivative as $Y/2$, matching the "half" on $T^a = \sigma^a/2$. Other authors absorb the $\frac{1}{2}$ into the definition of g'.

1.5 — The Gell-Mann–Nishijima relation (the bridge to QED)

Key Result

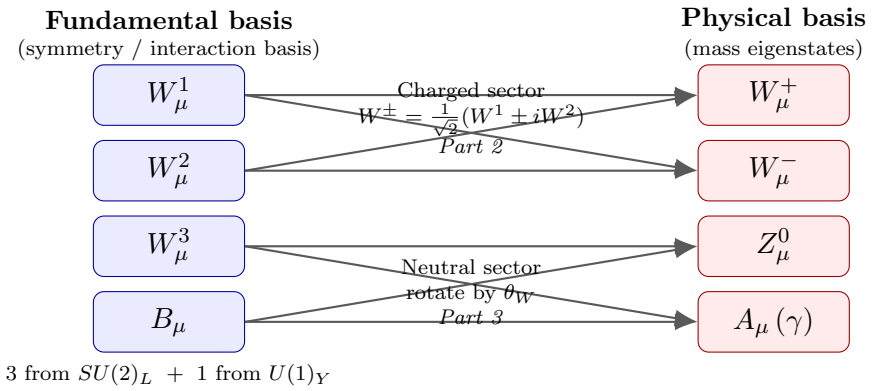
The Gell-Mann–Nishijima relation:

$$\boxed{Q \;=\; T^{(3)} \;+\; \tfrac{1}{2}\,Y}$$

where Q is electric charge, $T^{(3)}$ is the third component of weak isospin, and Y is weak hypercharge. This relation is *the* reason the electroweak theory "contains" QED.

Physical Insight

The Gell-Mann–Nishijima relation tells us that the electric charge Q is a *linear combination* of the diagonal $SU(2)_L$ generator ($T^{(3)}$) and the $U(1)_Y$ generator (Y). Consequence: the gauge boson that couples to Q (the photon) must be a *linear combination* of the gauge boson that couples to $T^{(3)}$ (which is W^3_μ) and the gauge boson that couples to Y (which is B_μ). **The mixing in this problem is already built into the Gell-Mann–Nishijima formula.**



Key Result

Part 1 summary. The electroweak gauge group $SU(2)_L \otimes U(1)_Y$ forces four gauge fields:

- Three from $SU(2)_L$: $W_\mu^1, W_\mu^2, W_\mu^3$, all with coupling g_W, coupling to weak isospin currents j_μ^a.
- One from $U(1)_Y$: B_μ with coupling g', coupling to weak hypercharge current j_μ^Y.

The Gell-Mann–Nishijima relation $Q = T^{(3)} + Y/2$ guarantees that some linear combination of W^3 and B couples to electric charge — i.e., that the photon is a mixture, not a pure W^3 or pure B. Finding that linear combination is what Parts 3–4 do.

Part 2 — Charged sector: $W^{(1,2)} \to W^{\pm}$

Problem Statement

Restatement. Express the physical *charged* bosons W_μ^+ and W_μ^- as linear combinations of W_μ^1 and W_μ^2. This part does *not* involve the weak mixing angle — it's pure $SU(2)$ algebra.

Deliverables

1. Explain why W^+ and W^- *must* be complex fields (not real ones).
2. Derive the Griffiths convention $W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \pm iW_\mu^2)$, including where the $\frac{1}{\sqrt{2}}$ comes from.
3. Show that this gives back the $V-A$ vertex factor $-i\,(g_W/\sqrt{2})\,\gamma^\mu \frac{1}{2}(1-\gamma^5)$ we already know.

2.1 — Why the charged sector mixes (a physical argument)

Step 2.1: Two real fields cannot represent a charged particle

A real-valued classical field $\phi(x) = \phi^*(x)$ corresponds, upon quantization, to a *neutral* particle that is its own antiparticle (like the π^0 or the photon). A *complex* field $\phi(x) \neq \phi^*(x)$ corresponds to a *charged* particle whose antiparticle is distinct (like the $\pi^\pm$).
The fields W_μ^1 and W_μ^2 are both real (they're the components of an $SU(2)$ gauge connection in the standard hermitian basis). So neither one alone can describe W^+ (which has $Q=+1$) or W^- (which has $Q=-1$). We need to combine them into a *complex* field whose conjugate is the antiparticle field.

Intuition

This is exactly the same trick that turns two real Klein-Gordon fields into one charged Klein-Gordon field. If ϕ_1, ϕ_2 are real with the same mass, then $\phi \equiv \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$ and $\phi^* \equiv \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2)$ are a charge-conjugate pair. This trick is so generic that it has nothing to do with the weak interaction per se — it's just *how you build a charged field from two neutral ones.*

2.2 — The $SU(2)$ raising/lowering operators

Step 2.2: Look at how W^1 and W^2 act inside the doublet

Inside an $SU(2)_L$ doublet $\chi_L = \binom{\nu_L}{e_L}$, the generators $T^a = \sigma^a/2$ act as

$$T^1 = \tfrac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T^2 = \tfrac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad T^3 = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Now the *raising* and *lowering* operators $T^\pm$ are defined exactly as in ordinary angular momentum:

$$T^\pm \equiv T^1 \pm iT^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

T^+ takes $e_L \to \nu_L$ (raises $T^{(3)}$), and T^- does the reverse.
The physical $W^\pm$ fields are exactly the components of the gauge field that couple to these raising/lowering operators.

Term Refresher

Why the analogy with $J_\pm$ in angular momentum? It's not analogy — it's the same mathematics. Weak isospin $\vec{T}$ obeys exactly the $SU(2)$ commutation relations $[T^a, T^b] = i\epsilon^{abc}T^c$, just like ordinary angular momentum. The operator T^+ raises $T^{(3)}$, T^- lowers it. Two states in a doublet have $T^{(3)} = \pm\frac{1}{2}$, exactly like a spin-$\frac{1}{2}$ particle. (You used this same SU(2) bridge in the QCD/SU(3) work earlier — the weak isospin is just the leftmost subgroup of the same mathematical structure.)

2.3 — Define $W^\pm$ to make the Lagrangian clean

Step 2.3: Combine W^1 and W^2 in the same way

Look at the charged piece of the gauge interaction (Griffiths Eq. 9.146–9.147):

$$\mathcal{L}_{\text{CC}} \;=\; -\,g_W\big(j^1_\mu W^{1\mu} + j^2_\mu W^{2\mu}\big)\,.$$

We *want* to write this as $-\frac{g_W}{\sqrt{2}}(j^+_\mu W^{-\mu} + j^-_\mu W^{+\mu})$. That requires defining

$$\boxed{W^\pm_\mu \;\equiv\; \frac{1}{\sqrt{2}}\big(W^1_\mu \pm i\,W^2_\mu\big) \quad \textbf{[Griffiths Eq. 9.148]}}$$

and the corresponding currents $j^\pm_\mu \equiv j^1_\mu \mp i j^2_\mu$.
Verification (one line of algebra).

$$\begin{aligned} j^+_\mu W^{-\mu} + j^-_\mu W^{+\mu} &= (j^1 - ij^2)_\mu \cdot \tfrac{1}{\sqrt{2}}(W^1 - iW^2)^\mu + (j^1 + ij^2)_\mu \cdot \tfrac{1}{\sqrt{2}}(W^1 + iW^2)^\mu \\ &= \tfrac{1}{\sqrt{2}}\big[(j^1W^1 + j^2W^2) - i(j^1W^2 - j^2W^1) + (j^1W^1 + j^2W^2) + i(j^1W^2 - j^2W^1)\big] \\ &= \tfrac{2}{\sqrt{2}}(j^1_\mu W^{1\mu} + j^2_\mu W^{2\mu}) \\ &= \sqrt{2}\,(j^1_\mu W^{1\mu} + j^2_\mu W^{2\mu}). \end{aligned}$$

The two cross-terms $\pm i(j^1W^2 - j^2W^1)$ cancel exactly — this is the whole point. So

$$j^1_\mu W^{1\mu} + j^2_\mu W^{2\mu} \;=\; \tfrac{1}{\sqrt{2}}\big(j^+_\mu W^{-\mu} + j^-_\mu W^{+\mu}\big)\,,$$

and therefore $\mathcal{L}_{\text{CC}} = -\frac{g_W}{\sqrt{2}}\big(j^+_\mu W^{-\mu} + j^-_\mu W^{+\mu}\big)$, which is the standard charged-current form.

Intuition

Where does the $\frac{1}{\sqrt{2}}$ come from? It's just normalization. We want the two real fields (W^1, W^2) to map onto a complex field $(W^+, W^{+*}) = (W^+, W^-)$ such that the kinetic term in the Lagrangian retains canonical normalization. Specifically, $-\frac{1}{4}\left[(F^1_{\mu\nu})^2 + (F^2_{\mu\nu})^2\right]$ becomes $-\frac{1}{2}F^+_{\mu\nu}F^{-\,\mu\nu}$ when you make the substitution with the $\frac{1}{\sqrt{2}}$ factor. Drop the $\frac{1}{\sqrt{2}}$ and you'd have to write $-\frac{1}{2}$ in some places and -1 in others.

2.4 — Recover the $V - A$ vertex factor

Step 2.4: Check that the standard vertex factor pops out

Consider the process $e^- \to \nu_e + W^-$ (or its crossed version). Griffiths (Eq. 9.149) writes the relevant current as

$$j_\mu^- \;=\; \bar{\nu}_L \gamma_\mu e_L \;=\; \bar{\nu}\,\gamma_\mu \tfrac{1}{2}(1-\gamma^5)\,e,$$

where the second form uses the chiral projector $P_L = \frac{1}{2}(1-\gamma^5)$. Plugging into $\mathcal{L}_{\text{CC}} \supset -\frac{g_W}{\sqrt{2}}\, j_\mu^- W^{+\mu}$:

$$\mathcal{L} \;\supset\; -\frac{g_W}{\sqrt{2}}\,\bar{\nu}\,\gamma_\mu \tfrac{1}{2}(1-\gamma^5)\,e\,W^{+\mu},$$

giving the vertex factor

$$\boxed{-\,i\,\frac{g_W}{\sqrt{2}}\,\gamma^\mu \tfrac{1}{2}(1-\gamma^5) \quad \text{(Griffiths Eq. 9.150 / Eq. 9.5)}}$$

which is precisely the $V - A$ vertex factor we started with way back in §9.1. Consistency check passed.

Warning / Convention Pitfall

Sign convention pitfall (this comes up in homework grading every semester). Two conventions exist for the sign in front of iW^2:

- **Griffiths** (Eq. 9.148): $W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \pm iW_\mu^2)$.
- **Thomson** (Eq. 15.12) and **Mann** (Eq. 23.45): $W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2)$.

The two conventions differ *only* in which combination is labeled "+" versus "−". The physics is identical. But you must be consistent — whatever sign you pick for $W^\pm$, you must use the same sign for $T^\pm = T^1 \pm iT^2$ and $j^\pm$. *Don't cross conventions across textbooks mid-derivation.* For Frantz's class, follow Griffiths' convention since the rest of the chapter does.

Key Result

Part 2 result.

$$\boxed{W_\mu^\pm \;=\; \frac{1}{\sqrt{2}}\big(W_\mu^1 \pm i\,W_\mu^2\big)} \qquad \text{(Griffiths Eq. 9.148; charged sector mixing).}$$

This is a unitary change of basis: two real fields (W^1, W^2) get repackaged as one complex field W^+ and its conjugate W^-. **No mixing angle, no Higgs, no electroweak unification needed.** It's just $SU(2)$ raising/lowering structure.

Part 3 — Neutral sector: $\{W^3, B\} \to \{\gamma, Z\}$

Problem Statement

Restatement. Express the physical *neutral* bosons — the photon A_μ and the neutral weak boson Z_μ — as linear combinations of the gauge fields W^3_μ and B_μ. This is where the weak mixing angle θ_W enters, and it's the heart of the question.

Deliverables

1. Argue why neither W^3 alone nor B alone can be the photon.
2. Argue why the most general transformation between two real, neutral fields with the same kinetic term is an $SO(2)$ rotation (one parameter, θ_W).
3. Write down the rotation in Griffiths' convention (Eq. 9.151).
4. Write the inverse relation $\{W^3, B\}$ in terms of $\{A, Z\}$.

3.1 — Why $W^3 \neq \gamma$

Step 3.1: W^3 couples only to left-handed fermions

The current that W^3 couples to is j^3_μ from the $SU(2)_L$ algebra:

$$j^3_\mu \;=\; g_W\,\bar{\chi}_L \gamma_\mu T^{(3)} \chi_L \;=\; \tfrac{1}{2} g_W (\bar{\nu}_L \gamma_\mu \nu_L - \bar{e}_L \gamma_\mu e_L)\,.$$

Notice everything here is left-handed (subscript L). Right-handed fermions are $SU(2)_L$ singlets and don't appear. **So W^3 couples to LH fermions only — it is maximally parity-violating.**
The photon, on the other hand, couples to electric charge Q, which is parity-conserving (QED preserves parity). Therefore $W^3 \neq \gamma$.

3.2 — Why $B \neq \gamma$

Step 3.2: B couples to weak hypercharge Y

The current that B couples to is j^Y_μ, summed over both chiralities with potentially different Y values:

$$j^Y_\mu \;=\; \sum_f Y_f\, \bar{\psi}_f \gamma_\mu \psi_f .$$

For the lepton doublet, the assignments are $Y_{e_L} = Y_{\nu_L} = -1$ and $Y_{e_R} = -2$, $Y_{\nu_R} = 0$ (Thomson §15.3 and Griffiths §9.7.2). These are *not* proportional to electric charge Q (else there would be no need for a separate hypercharge). For example, the neutrino has $Q = 0$ but $Y_{\nu_L} = -1 \neq 0$.
Therefore $B \neq \gamma$ either. **The photon must be a non-trivial linear combination of both W^3 and B.**

Physical Insight

The two arguments together rule out the "trivial" identifications and force the photon to be a real mixture. Looking at it through the Gell-Mann–Nishijima lens:

$$Q \;=\; T^{(3)} \;+\; \tfrac{1}{2}Y \quad\Rightarrow\quad \gamma \text{ couples to a combination of } T^{(3)} \text{ and } Y.$$

And since W^3 couples to $T^{(3)}$ and B couples to Y, the photon must be a combination of W^3 and B. The orthogonal combination is the Z^0.

3.3 — The most general orthogonal mixing in 2D is a rotation

Step 3.3: Write down the most general allowed mixing

We have two real, neutral fields W^3_μ and B_μ with canonically-normalized kinetic terms:

$$\mathcal{L}_{\text{kin}} \;\supset\; -\tfrac{1}{4}\,(F^{(W^3)}_{\mu\nu})^2 \;-\; \tfrac{1}{4}\,(F^{(B)}_{\mu\nu})^2.$$

Any change of basis $(W^3, B) \to (A, Z)$ that *preserves* the canonical normalization of the kinetic terms must be an orthogonal transformation. In 2D, the only orthogonal transformations are

- Rotations (determinant $+1$): $\binom{A}{Z} = R(\theta)\binom{B}{W^3}$ where $R(\theta) = \left(\begin{smallmatrix}\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{smallmatrix}\right)$.
- Reflections (determinant -1): same as a rotation followed by flipping a sign.

Reflections just amount to relabeling, so without loss of generality the mixing is a rotation by some angle θ_W:

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} \;=\; \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W^3_\mu \end{pmatrix}.$$

Reading off rows:

$$\boxed{\begin{aligned} A_\mu &= \;\; B_\mu \cos\theta_W \;+\; W^3_\mu \sin\theta_W, \\ Z_\mu &= -B_\mu \sin\theta_W \;+\; W^3_\mu \cos\theta_W \end{aligned}}$$

[**Griffiths Eq. 9.151**; Thomson Eqs. 15.17–15.18]

This is the central relation requested by H2.

Concept / Definition

What "mixing" means here. The fields $\{W^3, B\}$ are the gauge fields you'd write down if you were trying to make the gauge *Lagrangian* look as simple as possible (symmetry is manifest in this basis). The fields $\{A, Z\}$ are the ones that propagate as free particles with definite mass (mass eigenstates). These are two different orthogonal bases of the same 2-dimensional space of neutral fields, and they're related by a single rotation angle θ_W.
This is exactly analogous to:

- Neutrino flavor states (ν_e, ν_μ) vs. mass states (ν_1, ν_2) — mixing matrix is the PMNS rotation by angle θ_{12}.
- Quark flavor states (d, s) vs. mass states (real d, s quarks) — mixing matrix is the Cabibbo rotation by angle θ_C.

In each case: one basis is natural for writing the interaction Lagrangian, the other is natural for identifying physical particles. The rotation between them encodes physics; the *angle* is a measured parameter.

3.4 — The inverse relation

Since rotations are orthogonal, $R^{-1} = R^T$. Inverting:

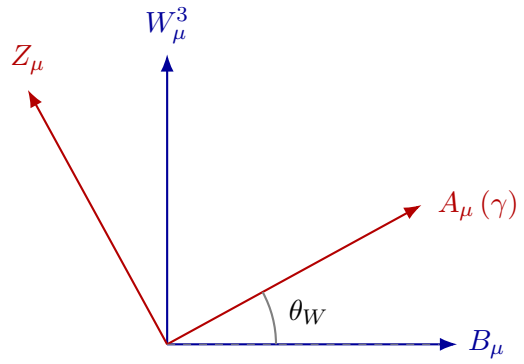
$$\begin{pmatrix} B_\mu \\ W^3_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix}, \qquad \boxed{\begin{aligned} B_\mu &= A_\mu \cos\theta_W - Z_\mu \sin\theta_W, \\ W^3_\mu &= A_\mu \sin\theta_W + Z_\mu \cos\theta_W. \end{aligned}} \tag{3}$$

We will need this inverse form in Part 4 when we substitute back into the gauge interaction Lagrangian to determine θ_W.

Warning / Convention Pitfall

Convention check vs. Mann. Mann (Eq. 23.45) writes the inverse rotation in the opposite direction, which corresponds to $\theta_W \to -\theta_W$. This is just a sign convention — the physical angle is the same. **Griffiths/Thomson/Halzen-Martin/Lancaster & Blundell all agree on the sign.** Mann is the outlier. Stick with Griffiths.

3.5 — Visual: the rotation in field space



The physical photon A_μ and Z_μ are obtained from the fundamental B_μ and W^3_μ by a rotation through $\theta_W \approx 28.7°$.

Intuition

The picture says: *the photon is mostly B with a $\sin^2\theta_W \approx 23\%$ admixture of W^3*. Concretely, since $\sin^2\theta_W \approx 0.231$:

- $A_\mu \approx 0.88\,B_\mu + 0.48\,W^3_\mu$ (the photon, massless).
- $Z_\mu \approx -0.48\,B_\mu + 0.88\,W^3_\mu$ (the Z^0, ~ 91 GeV).

The Z is the orthogonal combination — mostly W^3 with a smaller B component. This is why the Z couples to fermions in a more complicated way than the photon does (it has both vector and axial-vector pieces, with strengths that depend on θ_W).

Key Result

Part 3 result — the answer to H2.

$$\boxed{\begin{aligned} A_\mu &= \;\; B_\mu \cos\theta_W \;+\; W^3_\mu \sin\theta_W \quad (=\gamma), \\ Z_\mu &= -B_\mu \sin\theta_W \;+\; W^3_\mu \cos\theta_W . \end{aligned}}$$

θ_W is a single mixing angle (the Weinberg angle), measured experimentally.

Part 4 — Determining θ_W and the unification relation

Problem Statement

Restatement. The mixing angle θ_W is not arbitrary — it is fixed by demanding that one of the rotated fields (A_μ) couple with strength g_e to electric charge Q. Show how this requirement gives Griffiths' Eq. 9.154: $g_W \sin\theta_W = g' \cos\theta_W = g_e$, and equivalently $\tan\theta_W = g'/g_W$.

Deliverables

1. Substitute the inverse rotation $W^3 = A\sin\theta_W + Z\cos\theta_W$ and $B = A\cos\theta_W - Z\sin\theta_W$ into the neutral piece of the Lagrangian.
2. Read off the coefficient of A_μ and demand it equal $-g_e\, j^{\text{em}}_\mu$.
3. Use the Gell-Mann–Nishijima decomposition $j^{\text{em}} = j^3 + \frac{1}{2}j^Y$ to match coefficients of j^3 and j^Y separately.
4. Out drops the consistency relation and $\tan\theta_W = g'/g_W$.
5. Quote the experimental value of $\sin^2\theta_W$.

Strategy

The whole derivation is one substitution + one matching. The hardest part is keeping track of the factor of $\frac{1}{2}$ in front of g' (Griffiths' convention), which combines with the factor of $\frac{1}{2}$ from the Gell-Mann–Nishijima formula $Q = T^{(3)} + \frac{1}{2}Y$ to give the simple form of Eq. 9.154.

4.1 — Substitute the inverse rotation into the Lagrangian

Step 4.1: Plug in W^3 and B in terms of A and Z

The neutral piece of the gauge interaction Lagrangian (from Eq. 2) is

$$\mathcal{L}_{\text{neut}} \;=\; -\,g_W\, j^3_\mu W^{3\mu} \;-\; \tfrac{g'}{2}\, j^Y_\mu B^\mu .$$

Substitute $W^{3\mu} = A^\mu\sin\theta_W + Z^\mu\cos\theta_W$ and $B^\mu = A^\mu\cos\theta_W - Z^\mu\sin\theta_W$:

$$\begin{aligned}\mathcal{L}_{\text{neut}} = &-\,g_W\, j^3_\mu (A^\mu\sin\theta_W + Z^\mu\cos\theta_W) \\ &-\,\tfrac{g'}{2}\, j^Y_\mu (A^\mu\cos\theta_W - Z^\mu\sin\theta_W)\,. \end{aligned} \tag{4}$$

Now collect terms separately by which physical field A^μ or Z^μ they multiply:

$$\begin{aligned}\mathcal{L}_{\text{neut}} = &-\,A^\mu\Big(g_W\sin\theta_W\, j^3_\mu \;+\; \tfrac{g'}{2}\cos\theta_W\, j^Y_\mu\Big) \\ &-\,Z^\mu\Big(g_W\cos\theta_W\, j^3_\mu \;-\; \tfrac{g'}{2}\sin\theta_W\, j^Y_\mu\Big)\,. \end{aligned} \tag{5}$$

This is just the Lagrangian rewritten in the new basis. *Nothing about electroweak unification has been imposed yet — this is purely algebra.*

4.2 — The QED coupling sets the constraint

Step 4.2: Demand that the photon term has QED form

The standard QED interaction is

$$\mathcal{L}_{\text{QED}} \;=\; -\,g_e\, j^{\text{em}}_\mu A^\mu, \qquad j^{\text{em}}_\mu \;=\; \sum_f Q_f\, \bar\psi_f \gamma_\mu \psi_f .$$

For the electroweak theory to *contain* QED at low energies, the A_μ piece in (5) must equal $-g_e\, j^{\text{em}}_\mu A^\mu$. Therefore we demand:

$$g_W \sin\theta_W\, j^3_\mu \;+\; \tfrac{g'}{2}\cos\theta_W\, j^Y_\mu \;\overset{!}{=}\; g_e\, j^{\text{em}}_\mu .$$

4.3 — Use Gell-Mann–Nishijima to match coefficients

Step 4.3: Decompose j^{em} in terms of j^3 and j^Y

The Gell-Mann–Nishijima relation $Q = T^{(3)} + \frac{1}{2}Y$, written at the current level (integrate the operator equation against $\bar\psi\gamma_\mu\psi$), gives

$$j^{\text{em}}_\mu \;=\; j^3_\mu \;+\; \tfrac{1}{2}\, j^Y_\mu .$$

So the demand becomes

$$g_W \sin\theta_W\, j^3_\mu \;+\; \tfrac{g'}{2}\cos\theta_W\, j^Y_\mu \;=\; g_e\big(j^3_\mu \;+\; \tfrac{1}{2}\, j^Y_\mu\big) .$$

Match coefficients of j^3_μ and j^Y_μ separately (since they're independent currents):

$$[\text{coeff. of } j^3]\text{:} \quad g_W \sin\theta_W \;=\; g_e, \tag{6}$$

$$[\text{coeff. of } j^Y]\text{:} \quad \tfrac{g'}{2}\cos\theta_W \;=\; \tfrac{g_e}{2} \quad\Longleftrightarrow\quad g'\cos\theta_W \;=\; g_e. \tag{7}$$

4.4 — The unification consistency relation

Key Result

Combining the two matching equations:

$$\boxed{g_W \sin\theta_W \;=\; g' \cos\theta_W \;=\; g_e} \qquad \textbf{[Griffiths Eq. 9.154]}$$

Solve for θ_W by dividing:

$$\boxed{\tan\theta_W \;=\; \frac{g'}{g_W}}$$

Solve for the trig functions individually using $\sin^2 + \cos^2 = 1$:

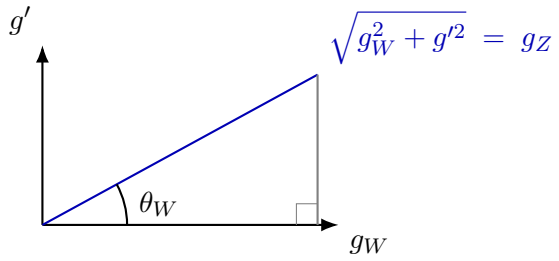
$$\sin\theta_W \;=\; \frac{g'}{\sqrt{g_W^2 + g'^2}}, \qquad \cos\theta_W \;=\; \frac{g_W}{\sqrt{g_W^2 + g'^2}}, \qquad g_e \;=\; \frac{g_W\, g'}{\sqrt{g_W^2 + g'^2}}.$$

Physical Insight

Take a moment to appreciate what just happened. We started with two independent gauge couplings g_W (for $SU(2)_L$) and g' (for $U(1)_Y$). We rotated W^3 and B by some angle θ_W. **Just by demanding that the resulting photon A_μ couple to electric charge with strength g_e, we got two equations in two unknowns (θ_W and the relation between g_W, g', g_e).** The result fixes *both* the angle and a non-trivial relation among the three couplings. This is what physicists mean when they say the electroweak theory "unifies" electromagnetism and the weak force.

The theory does *not* predict θ_W from first principles — you still have to measure either g_W and g' (or, equivalently, g_e and θ_W) from experiment. But it predicts a *relation* among them. Three couplings, two equations, one free parameter.

4.5 — The geometric picture: a right triangle



The triangle of electroweak couplings: legs g_W and g', hypotenuse $g_Z = \sqrt{g_W^2 + g'^2} = g_W / \cos\theta_W$, opposite/adjacent ratio $\tan\theta_W = g'/g_W$.

Intuition

Why is this triangle useful? It packages the three couplings g_W, g', g_e and the angle θ_W into a single geometric object. Every fact you'd want to remember about the electroweak couplings can be read off this triangle:

- $\tan\theta_W = g'/g_W$ (opposite over adjacent).
- $\sin\theta_W = g'/g_Z$, $\cos\theta_W = g_W/g_Z$ where $g_Z \equiv \sqrt{g_W^2 + g'^2}$.
- $g_e = g_W \sin\theta_W = g' \cos\theta_W$ (drop the perpendicular from the right angle to the hypotenuse; the foot of the perpendicular is at distance g_e from the origin in the appropriate direction).

The hypotenuse g_Z is the coupling of the Z^0 to fermions (modulo the c_V, c_A factors — see Griffiths Eq. 9.94 / Eq. 9.156).

4.6 — The Z^0 coupling drops out as a bonus

Step 4.6: Use the consistency relation to simplify the Z piece

Look at the Z^μ coefficient in Eq. (5):

$$\mathcal{L}_Z \;=\; -\,Z^\mu\Big(g_W\cos\theta_W\, j^3_\mu \;-\; \tfrac{g'}{2}\sin\theta_W\, j^Y_\mu\Big)\,.$$

Use $g' = g_e/\cos\theta_W$ and $g_W = g_e/\sin\theta_W$ to eliminate g_W and g':

$$\begin{aligned}
\mathcal{L}_Z &= -\,Z^\mu\left(\frac{g_e\cos\theta_W}{\sin\theta_W}\, j^3_\mu \;-\; \frac{g_e\sin\theta_W}{2\cos\theta_W}\, j^Y_\mu\right)\\
&= -\,\frac{g_e}{\sin\theta_W\cos\theta_W}\, Z^\mu\big(\cos^2\theta_W\, j^3_\mu \;-\; \tfrac{1}{2}\sin^2\theta_W\, j^Y_\mu\big)\\
&= -\,\frac{g_e}{\sin\theta_W\cos\theta_W}\, Z^\mu\big(j^3_\mu \;-\; \sin^2\theta_W\, j^{\rm em}_\mu\big)\,, \qquad (8)
\end{aligned}$$

where in the last line I used $j^{\rm em} = j^3 + \tfrac{1}{2}j^Y$, so $\cos^2\theta_W\, j^3 - \tfrac{1}{2}\sin^2\theta_W\, j^Y = j^3 - \sin^2\theta_W\,(j^3 + \tfrac{1}{2}j^Y) = j^3 - \sin^2\theta_W\, j^{\rm em}$.
Defining the Z^0 coupling

$$g_Z \;\equiv\; \frac{g_e}{\sin\theta_W\cos\theta_W} \;=\; \sqrt{g_W^2 + g'^2} \quad \text{(Griffiths Eq. 9.156)},$$

(8) reads

$$\mathcal{L}_Z \;=\; -\,g_Z\, Z^\mu\big(j^3_\mu - \sin^2\theta_W\, j^{\rm em}_\mu\big)\,.$$

This is Griffiths Eq. 9.155. The structure $j^3 - \sin^2\theta_W\, j^{\rm em}$ is exactly what generates the entries in Griffiths Table 9.1 (the famous c_V and c_A couplings of the Z^0 to each fermion species).

4.7 — Experimental value of θ_W

Concept / Definition

The Weinberg angle is *not* predicted by GWS; it must be measured. Modern global fits to LEP, SLD, Tevatron, and LHC data give

$$\sin^2\theta_W \approx 0.23146 \pm 0.00012 \quad \text{(Thomson Eq. 15.31; Mann Eq. 22.5)},$$

which corresponds to $\theta_W \approx 28.75°$. Griffiths quotes the same value (his Eq. 9.95 / Eq. 10.95) to slightly less precision: $\sin^2\theta_W \approx 0.23$, $\theta_W \approx 28.7°$.

Intuition

The fact that $\sin^2\theta_W$ is *neither* 0 *nor* 1 is what makes the electroweak theory *electroweak* — the photon and the Z are genuine *mixtures* of W^3 and B, not just one or the other:

- If $\theta_W = 0$: $\gamma = B$, $Z = W^3$. No mixing, two unrelated $U(1)$'s, no unification.
- If $\theta_W = 90°$: $\gamma = W^3$, $Z = -B$. Photon would couple only to LH fields (parity-violating EM!) — unphysical.
- Measured $\theta_W \approx 28.7°$: photon is $\sim 77\%$ B + $\sim 23\%$ W^3. The 23% admixture of W^3 is what carries the (electromagnetic-like) coupling of the photon to charged *left-handed* fermions through the weak isospin generator, completing the electroweak unification.

Key Result

Part 4 result.

$$\boxed{\tan\theta_W = \frac{g'}{g_W}, \qquad g_W \sin\theta_W = g'\cos\theta_W = g_e, \qquad g_Z = \frac{g_e}{\sin\theta_W\cos\theta_W} = \sqrt{g_W^2 + g'^2}.}$$

The single parameter θ_W ties together *three* couplings (g_W, g', g_e) and *four* gauge fields $(W^{1,2,3}, B \to W^\pm, Z, \gamma)$. Experimental: $\sin^2\theta_W \approx 0.231$.

Part 5 — *Why* does the mixing happen? Higgs mechanism

Problem Statement

Restatement. Conceptually: *why* are $\{W^{1,2,3}, B\}$ *not* themselves the observed bosons? What dynamical mechanism forces the rotation? Explain the role of the Higgs vacuum expectation value (VEV).

Deliverables

Give a one-page conceptual explanation: the Higgs VEV spontaneously breaks $SU(2)_L \otimes U(1)_Y \to U(1)_{\text{em}}$, giving masses to three linear combinations of the gauge fields $(W^{\pm}, Z)$ and leaving one combination (γ) massless. Show how diagonalizing the (W^3, B) mass matrix reproduces the rotation of Part 3 and gives $m_W = m_Z \cos\theta_W$.

Strategy

There are two pictures of why the mixing happens:

- **Symmetry-counting picture:** $SU(2)_L \otimes U(1)_Y$ has 4 generators. $U(1)_{\text{em}}$ has 1 generator. Therefore $4 - 1 = 3$ generators are "broken" by the Higgs VEV. By Goldstone's theorem, you'd get 3 massless Goldstone bosons; in a gauge theory they are "eaten" by 3 of the gauge fields, which become massive. The remaining 1 gauge field stays massless. That's the photon.
- **Mass-matrix picture:** Compute the gauge boson mass matrix from $|D_\mu \Phi|^2$ evaluated at the Higgs VEV. The (W^3, B) block is non-diagonal — diagonalizing it gives *exactly* the rotation we wrote down in Part 3, with $\tan\theta_W = g'/g_W$, and tells us $m_W = m_Z \cos\theta_W$ as a free bonus.

The mass-matrix picture is the more concrete one and gives explicit predictions, so we'll sketch it here.

5.1 — The Higgs doublet and its VEV

Concept / Definition

The Higgs scalar Φ is an $SU(2)_L$ doublet with hypercharge $Y_\Phi = +1$:

$$\Phi(x) \;=\; \begin{pmatrix} \phi^+(x) \\ \phi^0(x) \end{pmatrix}, \qquad \langle \Phi \rangle \;=\; \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \qquad v \approx 246 \text{ GeV}.$$

The vacuum expectation value (VEV) v is the value of the lower component at the minimum of the Higgs potential $V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \frac{\lambda}{2}(\Phi^\dagger \Phi)^2$ with $\mu^2 > 0$. Because the VEV picks out a specific direction in the $SU(2)_L$ doublet space, it "breaks" the symmetry: the vacuum is not invariant under all of $SU(2)_L \otimes U(1)_Y$ — only under the subgroup that leaves $\langle \Phi \rangle$ invariant.

Which subgroup leaves $\langle \Phi \rangle$ invariant? A generator X leaves the vacuum invariant iff $X\langle \Phi \rangle = 0$. Plugging in $X = T^a$ $(a = 1, 2, 3)$ and $X = Y/2$, you find that $T^1, T^2, (T^3 - Y/2)$ all act non-trivially on $\langle \Phi \rangle$ (broken), while $(T^3 + Y/2)\langle \Phi \rangle = 0$ *exactly* because of the Gell-Mann–Nishijima formula $Q = T^{(3)} + Y/2$ and the fact that the lower component of Φ has $T^{(3)} = -1/2, Y = +1$ giving $Q = 0$. So Q generates the unbroken subgroup, $U(1)_{\text{em}}$. This is the photon's home.

5.2 — Compute the gauge field mass matrix

Step 5.2: Plug $\langle \Phi \rangle$ into the kinetic term

The Higgs kinetic term is $|D_\mu \Phi|^2$, with

$$D_\mu \Phi \;=\; \Big(\partial_\mu - i g_W \, T^a W_\mu^a - i \tfrac{g'}{2} \, Y_\Phi B_\mu \Big) \Phi.$$

At the VEV (replace $\Phi \to \langle \Phi \rangle$ and $\partial_\mu \to 0$):

$$D_\mu \langle \Phi \rangle \;=\; -i \Big(g_W \, T^a W_\mu^a + \tfrac{g'}{2} \, B_\mu \Big) \langle \Phi \rangle.$$

Computing $|D_\mu \langle \Phi \rangle|^2 = (D_\mu \langle \Phi \rangle)^\dagger (D^\mu \langle \Phi \rangle)$ and keeping the gauge-field bilinears, after some algebra one finds (Lancaster & Blundell Eq. 47.33, Mann Eq. 23.41-44):

$$|D_\mu \langle \Phi \rangle|^2 \;=\; \frac{v^2}{8} \left[g_W^2 (W_\mu^1)^2 + g_W^2 (W_\mu^2)^2 + (g_W W_\mu^3 - g' B_\mu)^2 \right]. \tag{9}$$

5.3 — Read off the masses

Step 5.3: Identify mass terms

Recall the canonical mass term for a vector field is $\frac{1}{2}m^2 V_\mu V^\mu$. Compare to (9):

- W^1 **and** W^2**:** each gets a mass term $\frac{g_W^2 v^2}{8}(W_\mu^a)^2 = \frac{1}{2}\cdot\frac{g_W^2 v^2}{4}(W_\mu^a)^2$, giving $m_{W^1} = m_{W^2} = \frac{g_W v}{2}$. After the $W^\pm = \frac{1}{\sqrt{2}}(W^1 \pm iW^2)$ rotation (Part 2), the same mass becomes

$$\boxed{m_W \;=\; \tfrac{1}{2}\, g_W\, v.}$$

- (W^3, B) **sector:** the term $(g_W W^3 - g' B)^2$ gives a 2×2 quadratic form in (W^3, B). Expanding,

$$(g_W W^3 - g' B)^2 \;=\; g_W^2 (W^3)^2 - 2 g_W g' W^3 B + g'^2 B^2.$$

  Writing this as $\frac{1}{2}V_\mu^T M^2 V^\mu$ with $V = (W^3, B)^T$:

$$M^2 \;=\; \frac{v^2}{4}\begin{pmatrix} g_W^2 & -g_W g' \\ -g_W g' & g'^2 \end{pmatrix}.$$

This mass matrix is *not* diagonal — the off-diagonal $-g_W g'$ entries are exactly what *forces* the rotation between $\{W^3, B\}$ and the mass eigenstates.

5.4 — Diagonalize the (W^3, B) mass matrix

Step 5.4: Find eigenvalues and eigenvectors

The mass matrix M^2 is rank-1 (its determinant is $g_W^2 g'^2 - g_W^2 g'^2 = 0$). Therefore it has one zero eigenvalue and one non-zero eigenvalue.

Trace gives the sum of eigenvalues:

$$\mathrm{tr}(M^2) \;=\; \tfrac{v^2}{4}(g_W^2 + g'^2) \;=\; \lambda_+ + \lambda_-.$$

Determinant gives the product:

$$\det\big(M^2\big) \;=\; 0 \;=\; \lambda_+\lambda_- \quad\Longrightarrow\quad \lambda_- = 0.$$

So

$$\lambda_+ \;=\; \tfrac{v^2}{4}(g_W^2 + g'^2), \qquad \lambda_- \;=\; 0.$$

Eigenvector for $\lambda_- = 0$ (the photon): solve $M^2\binom{a}{b} = 0$. The first row gives $g_W^2 a - g_W g' b = 0 \Rightarrow b = g_W a/g'$. Normalizing $a^2 + b^2 = 1$:

$$\begin{pmatrix} a \\ b \end{pmatrix} \;=\; \frac{1}{\sqrt{g_W^2 + g'^2}} \begin{pmatrix} g' \\ g_W \end{pmatrix} \;=\; \begin{pmatrix} \sin\theta_W \\ \cos\theta_W \end{pmatrix},$$

using $\sin\theta_W = g'/\sqrt{g_W^2 + g'^2}$ and $\cos\theta_W = g_W/\sqrt{g_W^2 + g'^2}$. So the massless eigenstate is

$$A_\mu \;=\; W_\mu^3 \sin\theta_W + B_\mu \cos\theta_W \;\overset{!}{=}\; \text{Griffiths Eq. 9.151 photon definition. } \checkmark$$

Eigenvector for λ_+ (the Z): the orthogonal direction:

$$Z_\mu \;=\; W_\mu^3 \cos\theta_W - B_\mu \sin\theta_W \;\overset{!}{=}\; \text{Griffiths Eq. 9.151 } Z \text{ definition. } \checkmark$$

Z mass:

$$\tfrac{1}{2} m_Z^2 \;=\; \lambda_+ \;=\; \tfrac{v^2}{4}(g_W^2 + g'^2) \quad\Longrightarrow\quad \boxed{m_Z \;=\; \tfrac{v}{2}\sqrt{g_W^2 + g'^2} \;=\; \frac{m_W}{\cos\theta_W}.}$$

Physical Insight

This is what "the mixing happens" actually means. Diagonalizing the gauge boson mass matrix *is* the rotation of Part 3. The Weinberg angle is the diagonalization angle of the (W^3, B) mass matrix, which is in turn fixed by the ratio of the two gauge couplings g'/g_W. The Higgs doesn't *predict* θ_W — the inputs g_W, g' are still independent — but it *causes* the rotation by giving an off-diagonal mass matrix.

The crucial fact is that the mass matrix has a *zero* eigenvalue. This is not an accident: it follows from the fact that the Higgs VEV is electrically neutral, so $Q\langle\Phi\rangle = 0$, so the gauge field that couples to Q stays massless. **The photon is the gauge field of the unbroken subgroup** $U(1)_{\mathrm{em}}$, exactly as required for QED to emerge at low energies.

5.5 — $m_W = m_Z \cos\theta_W$: a precision-test prediction

Key Result

From the diagonalization:

$$m_W = \tfrac{1}{2} g_W v, \qquad m_Z = \tfrac{v}{2}\sqrt{g_W^2 + g'^2} = \frac{m_W}{\cos\theta_W}.$$

This gives the famous tree-level prediction

$$\boxed{m_W = m_Z \cos\theta_W}$$ (Griffiths Eq. 9.98 / 10.98; Thomson §16.4).

With measured $m_Z = 91.188$ GeV and $\sin^2\theta_W \approx 0.231$ (so $\cos\theta_W \approx 0.877$), this predicts $m_W^{\text{pred}} \approx 79.94$ GeV, vs. measured $m_W = 80.385$ GeV. The small discrepancy is one of the most precisely-tested radiative-correction predictions of the Standard Model (top-quark and Higgs-mass loops; Thomson §16.4).

Intuition

One-sentence summary of Part 5. The Higgs vacuum is electrically neutral, so it "feels" the three broken generators $T^1, T^2, (T^3 - \frac{1}{2}Y)$ but *not* the unbroken combination $Q = T^{(3)} + \frac{1}{2}Y$. Three gauge fields couple to broken generators and become massive ($W^\pm, Z$); the fourth, which couples to Q, stays massless — and *that* is the photon.

Final Summary — the answer to H2 in one box

Key Result

Electroweak gauge group: $SU(2)_L \otimes U(1)_Y$.
Four "fundamental" gauge fields:

$$\underbrace{W_\mu^1,\, W_\mu^2,\, W_\mu^3}_{SU(2)_L \text{ triplet, coupling } g_W} \qquad \text{and} \qquad \underbrace{B_\mu}_{U(1)_Y \text{ singlet, coupling } g'} .$$

Four physical bosons (mass eigenstates):

$$\begin{aligned} W_\mu^{\pm} &= \tfrac{1}{\sqrt{2}}\left(W_\mu^1 \pm i\, W_\mu^2\right) && \text{(Griffiths Eq. 9.148)}, \\ A_\mu &= \;\; B_\mu \cos\theta_W \;+\; W_\mu^3 \sin\theta_W && \text{(Griffiths Eq. 9.151)}, \\ Z_\mu &= -B_\mu \sin\theta_W \;+\; W_\mu^3 \cos\theta_W && \text{(Griffiths Eq. 9.151)}. \end{aligned}$$

The Weinberg angle:

$$\tan\theta_W \;=\; \frac{g'}{g_W}, \qquad g_W \sin\theta_W \;=\; g' \cos\theta_W \;=\; g_e \quad \text{(Griffiths Eq. 9.154)}.$$

Why the mixing: the Higgs VEV $\langle \Phi \rangle = (0, v/\sqrt{2})^T$ spontaneously breaks $SU(2)_L \otimes U(1)_Y \to U(1)_{\text{em}}$. The off-diagonal (W^3, B) mass matrix is diagonalized by exactly the θ_W rotation above. The massless eigenstate is the photon; the orthogonal massive combination is the Z^0, with $m_W = m_Z \cos\theta_W$.
Experimental input: $\sin^2\theta_W \approx 0.231$, so $\theta_W \approx 28.7°$.

Quick-reference one-line cheat sheet

$$W^{\pm} = \tfrac{1}{\sqrt{2}}(W^1 \pm iW^2), \quad \begin{pmatrix} A \\ Z \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B \\ W^3 \end{pmatrix}, \quad \tan\theta_W = g'/g_W, \quad m_W = m_Z \cos\theta_W .$$